Hao-Xuan Sun

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EMPLOYMENT

School of Mathematics &
International Center for Interdisciplinary Statistics — **HARBIN INSTITUTE OF TECHNOLOGY**
Tenure-track Associate Professor — Jul. 2025 - now

School of Statistics — **LONDON SCHOOL OF ECONOMICS AND POLITICAL SCIENCE**
Visiting Researcher — Sep. 2026 - Dec. 2026 (expected)

EDUCATION

Academy for Advanced Interdisciplinary Studies — **PEKING UNIVERSITY**
PhD in Statistics, supervised by Prof. Song Xi Chen — Sep. 2019 - Jun. 2025

School of Arts — **PEKING UNIVERSITY**
B.S. in Arts (Double Degree) — Sep. 2016 - Jun. 2019

School of Mathematical Sciences — **PEKING UNIVERSITY**
B.S. in Statistics — Sep. 2015 - Jun. 2019

RESEARCH INTEREST

High-dimensional Statistics, Data Assimilation, Astrostatistics, Statistical Applications in Ocean Science

CITATION PROFILE

H-Index 6, **I-10 Index** 4, **Citations** 119 (Web of Science, without self citations); 242 (Google Scholar)

PUBLICATIONS

* Indicating co-first author

- **Sun, H-X.**, Chen, S.X. and Qiu, Y. (2026+) Localization Estimator for High-dimensional Tensor Covariance Matrices. *The Annals of Statistics*, to appear.
- Wang, Z., **Sun, H-X.**, and Jing, Z. (2026) Temporal heterogeneity of in situ ocean observing capacity could cause an artificial intensification of extreme warm water events globally. *Geophysical Research Letters*, 53, e2026GL123043.
- **Sun, H-X.**, Wang, S., Zheng, X. and Chen, S.X. (2024) High-dimensional Ensemble Kalman Filter with Localization, Inflation and Iterative Updates. *Quarterly Journal of the Royal Meteorological Society*. 150(765), 4870–4884.
- Xu, N., Hu, M., Li, X., Song K., Qiu Y., **Sun, H-X.**, Wang, Y., Zeng, L., Li, M., Wang, H., Hu, S., Zong, T., Bai, Y., Zhang, Z., Li, S., Shuai, S., Chen, Y. and Guo, S. (2024) Resolving Ultraviolet-visible Spectra for Complex Dissolved Mixtures of Multitudinous Organic Matters in Aerosols. *Analytical Chemistry*. 96 (5), 1834-1842.
- Huang, Y., Guo, B., **Sun, H-X.**, Liu, H. and Chen. S.X. (2021) Relative Importance of Meteorological Variables on Air Quality and Role of Boundary Layer Height. *Atmospheric Environment*. 267: 118737.
- Yan, H.*, Zhu, Y.*, Gu, J.*, Huang, Y.*, **Sun, H-X.***, Zhang, X., Wang, Y., Qiu, Y., Chen, S.X. (2021) Better Strategies for Containing COVID-19 Pandemic – A Study of 25 Countries via a vSIADR Model. *Proceeding of the Royal Society A*. 477(2248): 20200440.
- Gu, J.*, Yan, H.*, Huang, Y.*, Zhu, Y.*, **Sun, H-X.***, Qiu, Y. and Chen, S.X. (2020). Comparing Containment Measures among Nations by Epidemiological Effects of COVID-19. *National Science Review*. 7(12): 1847-1851.
- **Sun, H-X.**, Qiu, Y., Yan, H., Huang, Y., Zhu, Y., Gu, J. and Chen, S.X. (2020). Tracking Reproductivity of COVID-19 Epidemic in China with Varying Coefficient SIR Model (with discussion). *Journal of Data Science*. 18(3): 455-472.

MANUSCRIPT

† Indicating alphabetical order

- Chen, S.X.†, **Sun, H-X.**† and Wang, S.† (2026+) High Dimensional Ensemble Kalman Filter. *Major Revision at Bernoulli*.
- Cao, Y., Bai, J., Xu, R., Jin, Z., Wu, J., **Sun, H-X.**, Zhu, R., Lin, D., Wang, Y., Zhou, T. and Liu, K. (2026+) Facilitating Sustainable Shale Oil Exploration Through Transfer Learning. *Under review*.
- **Sun, H-X.** and Ye, Y. (2026+) Modified Cholesky Decomposition-Based Localization for Ensemble Kalman Filter. *Manuscript*.

TECHNICAL REPORTS

- Li, H., Liu, J., Wang, J., Zhan, H., Chen, H., Bian, M., Feng, Y., Yang, Q., Yan, X., Peng, A., **Sun, H-X.**, and Chen, S.X. (2024). Air Quality Assessment Report XI: Regional Air Quality Assessment of the 3+130 Cities. Center for Statistics at Peking University.
- Luo, S., Chen, H., Wang, J., Zhan, H., Lin, H., Liu, J., Ding, J., Wang, B., Yan, X., Dai, H., **Sun, H-X.**, Guo, B. and Chen, S.X. (2023). Air Quality Assessment Report X: Regional Air Quality Assessment of the 3+110 Cities. Center for Statistics at Peking University.
- **Sun, H-X.**, Luo, S., Zhang, X., Zhan, H., Lin, H., Chen, H., Lai, H., Tong, P., Guo, B. and Chen, S.X. (2022). Air Quality Assessment Report IX: Regional Air Quality Assessment of the 3+99 Cities. Center for Statistics at Peking University.
- **Sun, H-X.**, Luo, S., You, W., Wang, Y., Zhan, H., Wang, X., Huang, Y., He, J., Guo, B. and Chen, S.X. (2021). Air Quality Assessment Report VIII: Regional Air Quality Assessment of the 3+95 Cities. Center for Statistics at Peking University.
- **Sun, H-X.**, You, W., Guo, B., Luo, S., Zheng, X., Han, T., Wang, X., Liu, R., Huang, Y., He, J., Chen, L., Wang, H. and Chen, S.X. (2020). Air Quality Assessment Report VII: Regional Air Quality Assessment of the 2+66 Cities. Center for Statistics at Peking University.
- **Sun, H-X.**, Guo, B., Wang, X., Huang Y. and Chen, S.X. (2019). Assessment of Air Quality in Shandong Province from 2014 to 2018. Center for Statistics at Peking University.
- Liu, H., Ye, F., **Sun, H-X.**, Guo, B., Liu R., Xiao, J., He, J., Zheng, X., Wang, H. and Chen, S.X. (2019). Air Quality Assessment Report VI: Regional Air Quality Assessment of the 2+43 Cities. Center for Statistics at Peking University.

RESEARCH SUPPORT

Localization theory and algorithm for high-dimensional ensemble Kalman filter. **Jan. 2027-Dec. 2029**
National Natural Science Foundation of China. PI. RMB 300,000.

High-dimensional Ensemble Kalman Filter Methods and Algorithms. **Jan. 2026-Dec. 2028**
HIT Shenzhou Young Scholars Start-up Funding. PI. RMB 1,000,000.

Artificial Intelligence-Guided Methods for Shale Oil Development **Oct. 2026-Oct. 2027**
China National Petroleum Corporation Collaboration Project. Co-PI. RMB 500,000.

South China Sea Research Voyage **Aug. 2025-Sep.2025**
Scientific Researcher on board

Integrated Research on Multi-spheric Interactive Data Products in West Pacific **Jan. 2023-Dec. 2026**
National Natural Science Foundation of China. No.92358303, contributed

Mathematical Problems and Algorithms in Atmospheric System Modeling by Integrating Massive Observations **Jan. 2023-Dec. 2027**
National Natural Science Foundation of China. No.12292983, contributed

Statistical Inference for Varying Coefficient Epidemiological Models **Jan. 2021-Dec. 2024**
National Natural Science Foundation of China. No.12071013, participant (7/7)

Statistical Models for Air Quality Assessment, **Jan. 2019-Dec. 2020**
China National Key Research Fund. 2016YFC0207701, contributed

GRADUATE STUDENTS

Current Ph.D. students

Gao, Jiajia, started in 2026

Ye, Yuwei, started in 2026

Current Master students

Wang, Xueshuo, started in 2026

Zhou, Yuhan, started in 2026

Zhu, Qianli, started in 2026, co-supervised with Prof. Jin, Shuilin

AWARDS AND HONORS

- 2026 JCSDS Young Researcher Travel Award July 2026
- Academic Excellence Award, Peking University Oct 2024

- Merit Student of School, Peking University — Oct 2020
- Academic Scholarship, Peking University — Oct 2020
- Merit Graduate of School of Mathematical Sciences, Peking University — Jul 2019
- Yizheng Scholarship, Peking University — Sep 2017

TALKS

8th WMO Symposium on Data Assimilation & 12th International Symposium on Data Assimilation — **Nanjing, China**
Modified Cholesky Decomposition-Based Localization for Ensemble Kalman Filter — Aug. 2026

Departmental seminar at National University of Singapore — **Singapore, Singapore**
High-Dimensional Ensemble Kalman Filter with Covariance Localization Estimator — Aug. 2026

4rd Joint Conference on Statistics and Data Science (JCSDS) — **Guiyang, China**
Modified Cholesky Decomposition-Based Localization for Ensemble Kalman Filter — Jul. 2026

2026 JCSDS Young Researchers Workshop — **Guiyang, China**
Localization Estimator for High-dimensional Tensor Data — Jul. 2026

2025 International Chinese Statistical Association (ICSA) China Conference — **Shenzhen, China**
Localization Estimator for High-dimensional Tensor Data — Jun. 2026

1st Symposium on Ocean Statistics and Data Science — **Xiamen, China**
Covariance Estimation of High-Dimensional Ocean Tensor Data — Jan. 2026

17th Symposium on Ocean Data Assimilation and Numerical Modeling — **Guangzhou, China**
High-dimensional Ensemble Kalman Filter and Localization Estimator — Nov. 2025

11th International Symposium on Data Assimilation (ISDA) — **Melbourne, Australia**
High-dimensional Ensemble Kalman Filter and Localization Estimator — Oct. 2025

3rd Joint Conference on Statistics and Data Science (JCSDS) — **Hangzhou, China**
Localization Estimator for High-dimensional Tensor Data — Jul. 2025

2025 International Chinese Statistical Association (ICSA) China Conference — **Zhuhai, China**
High-dimensional Ensemble Kalman Filter with Localization, Inflation and Iterative Update — Jun. 2025

13th China National Conference on Probability and Statistics — **Xiamen, China**
Ocean Data Assimilation System — Nov. 2024

Workshop on Digital Twin and Uncertainty Quantification (DTUQ) — **Kunming, China**
High-dimensional Ensemble Kalman Filter with Localization, Inflation and Iterative Update — Aug. 2024

2nd Joint Conference on Statistics and Data Science (JCSDS) — **Kunming, China**
High-dimensional Ensemble Kalman Filter with Localization, Inflation and Iterative Update — Jul. 2024